

TECHNICAL & COMMERCIAL PROPOSAL

185.4 kWp Rooftop Solar PV System

299 × 620 Wp solar panels | 2 × 80 kW grid-tied inverters

PREPARED FOR

Kampala Marriott Hotel

Mezzanine roof, Kampala, Uganda

PREPARED BY

NileSun Power Ltd

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Ref: NSP-261105-KMH | 05-Nov-2026

Valid until 05-Dec-2026 | Clean Energy - Brighter Tomorrow

185.4 kWp Grid-Tied Rooftop Solar PV System

Quote & Client Details

Quote No.	NSP-261105-KMH	Date	05-Nov-2026	Valid Until	05-Dec-2026
Client Name	Kampala Marriott Hotel	Contact No.	Mr. Jagath +256 791 811805	Client Email	
Site Address	Kampala Marriott Hotel, mezzanine roof, Kampala	Supply Type	Three-phase 400 V, UEDCL, main LV switchboard	VAT TIN No.	

185.4 kWp Solar array capacity	**160 kW** Inverters (2 × 80 kW)	**299 nos** 620 Wp solar panels	**USD 102,789** Turnkey price excl. VAT
797 kWh Avg. daily generation	**291.1 MWh** Est. annual generation	**USD 25,343** Savings in year 1	**4.0 years** Payback period

1. Executive Summary

NileSun Power Ltd is pleased to submit this proposal for the design, supply, installation and commissioning of a **185.4 kWp grid-tied rooftop solar PV system** on the mezzanine roof of the **Kampala Marriott Hotel**: **299 × 620 W N-type bifacial solar panels** on a ballasted aluminium structure at 10° tilt, with **two 80 kW three-phase string inverters** connected to the hotel's main LV switchboard.

The array generates about **291,051 kWh per year** (approx. 797 kWh/day). The hotel used 981,901 kWh in October 2026 (UEDCL bill), so every solar kWh is used on site and directly reduces grid purchases. At the UEDCL shoulder tariff of UGX 357/kWh (≈ USD 0.0871/kWh at UGX 4,100/USD) this saves about **USD 25,343 in the first year**. The turnkey price is **USD 102,789** (excl. VAT), paid back in about **4.0 years**, with an IRR of **25.9 %** (Sections 12 and 14).

Key Benefits

- Lower bills: solar energy covers part of the hotel's daytime load at USD 0.040/kWh versus USD 0.087/kWh from the grid.
- Fast payback (4.0 years) and USD 763,309 savings over 25 years.
- Long life: Astronergy N-type bifacial panels (15-year product, 30-year power warranty); remote monitoring of generation and savings.

2. Equipment Schedule (Bill of Materials)

#	Item	Specification	Qty	UoM
1	Solar panels	Astronergy ASTRO N7 CHSM66RN(DG)/F-BH 620 Wp N-type TOPCon bifacial, 2382 × 1134 × 30 mm (Annex 4)	299	pcs
2	String inverters	Sigenergy Sigen PV 80M1, 80 kW three-phase, 1100 V DC, with monitoring (Annex 3)	2	pcs
3	Mounting structure	Enerack ballasted aluminium system, 10° tilt, 6 × 2P + 1 × 1P tables, no roof penetration (Annex 5)	1	system
4	DC protection	DC isolator units (2-pole 1000 V 32 A) + DC SPD Type II – 1 each per string	16+16	pcs
5	AC combiner panel	MCCBs 2 × 160 A (inverter incomers), 400 A roof isolator, AC SPD Type II	1	set
6	Grid connection	Connection to the existing Masterpact NW10 H1 solar incomer on the main LV switchboard	1	set
7	Cabling	Solar cable 4 mm², AC 4C × 50 mm² and 4C × 240 mm² Cu XLPE/SWA, cable tray	1	lot
8	Monitoring & control	Smart data logger, energy meter, remote monitoring	1	set
9	Earthing	Array bonding, connection to building earth and lightning protection	1	set

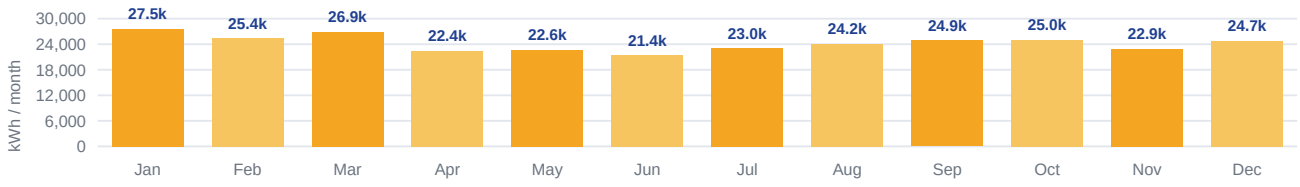
3. Solar Array & Inverters

Solar array	Value
Panels	299 × 620 W = 185.4 kWp
Arrangement	7 tables: 6 × 2P×23, 1 × 1P×23 (portrait)
DC / AC ratio	185.4 kWp / 160 kW = 1.16
Location / tilt	Flat concrete mezzanine roof, 10°
Panel area	approx. 808 m² (299 panels)

Inverters	Value
INV-01	152 panels = 94.2 kWp, 8 strings × 19
INV-02	147 panels = 91.1 kWp, 3 × 19 + 5 × 18
AC output	2 × 80 kW, 400 V 3N~
Connection	AC combiner → MLVS via riser

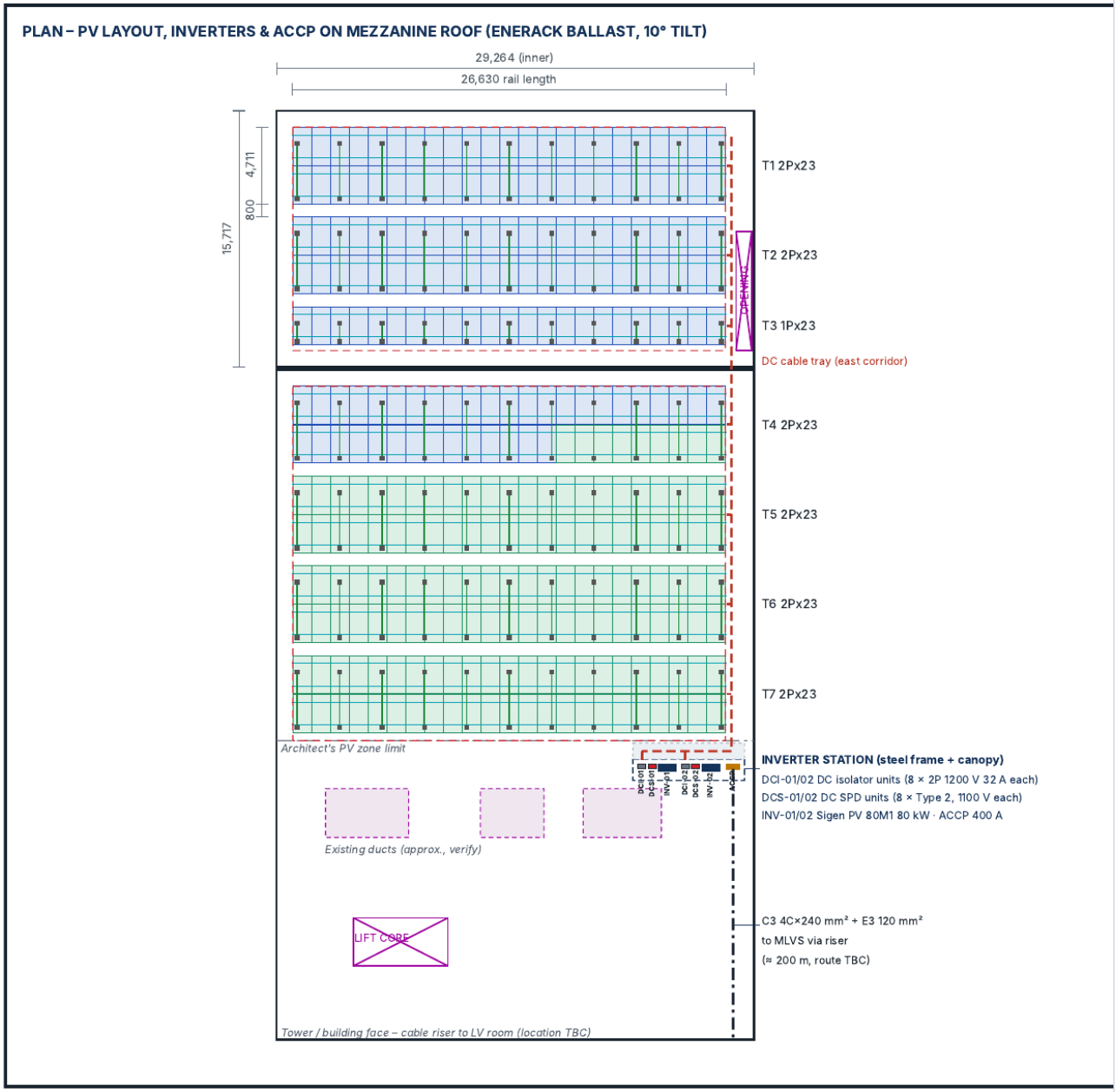
4. Estimated Energy Generation

PVsyst V8.1.8 simulation (P50, Meteonorm 9.0, PR 85.4 %, Annex 7). Annual total **291,051 kWh** (≈ 1,570 kWh/kWp).



5. Design Solar Top Installation (Roof Plan)

The 299 panels are placed on the mezzanine roof on 7 Enerack ballast tables at 10° tilt with 800 mm service aisles. INV-01 feeds tables T1–T4 (part), INV-02 feeds T4 (part)–T7. The inverter station and AC combiner panel sit at the south-east corner; the AC cable runs to the main LV switchboard via the building riser.



Proposed solar installation – Kampala Marriott Hotel, mezzanine roof (0°18'11.34"N, 32°35'13.76"E). Full drawing in Annex 2.

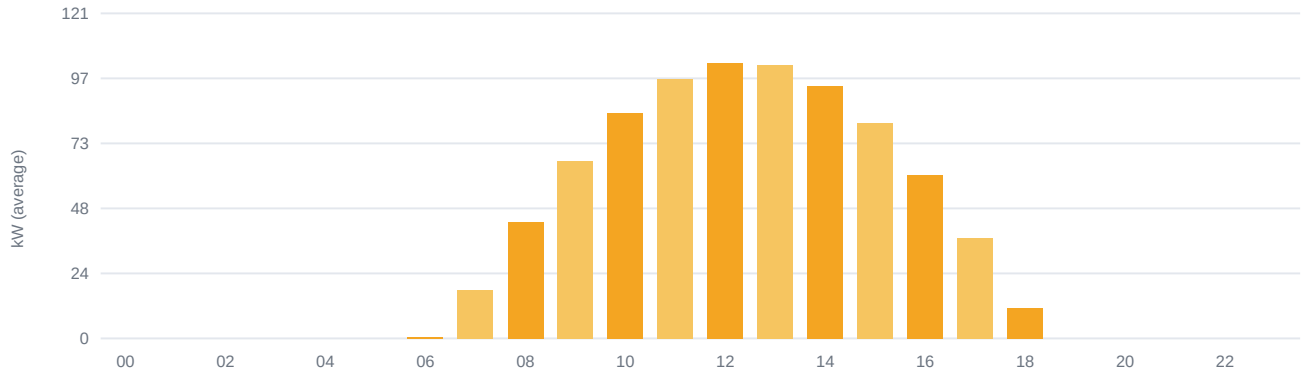
7. Hotel Load & Solar Contribution

The hotel used **981,901 kWh in October 2026** (UEDCL bill; ≈ 11.8 GWh per year), with an average daytime demand of about 1,348 kW and a maximum demand of 1,836 kW. The solar system supplies about **2.5 %** of this, so all solar output is consumed on site even on low-occupancy days – no export and no battery are needed.

Item	Value	Check	Result
Hotel annual consumption	11,782,812 kWh	Solar share of consumption	2.5 %
Solar generation, year 1	291,051 kWh	Peak solar output	≈ 103 kW (midday average)
Hotel daytime demand	far above solar output	Energy exported	0 kWh (all used on site)
Grid tariff (avoided)	USD 0.0871/kWh (UGX 357), +2 %/yr	Every solar kWh used?	Yes – offsets grid purchase

8. Solar Generation by Time-of-Use Band

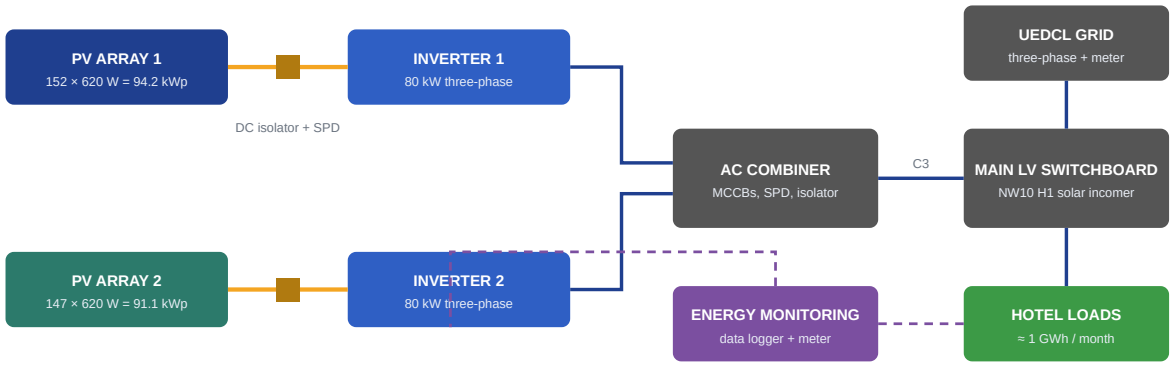
Average daily solar output by hour. All solar generation is counted in the shoulder period (06:00–18:00), when the hotel's air-conditioning, kitchen and laundry loads are high; no solar energy is assigned to the peak band.



TOU band	Hours	Hotel consumption (kWh/yr)	Solar energy (kWh/yr)	Share of solar	Reduction of band purchase
Off-peak	00:00–06:00	2,764,833	0	0.0 %	0.0 %
Shoulder	06:00–18:00	6,015,570	291,051	100.0 %	4.8 %
Peak	18:00–24:00	3,002,410	0	0.0 %	0.0 %
Total		**11,782,812**	**291,051**	**100 %**	**2.5 %**

*Consumption and tariffs from the UEDCL bill for October 2026 (tariff 30.2 Large Consumers); annual = October × 12. Rates are as billed (peak 428.8, shoulder 357.0, off-peak 286.5 UGX/kWh) at UGX 4,100/USD, excluding VAT; solar does not reduce the maximum-demand charge.

9. System Connection Diagram



10. Implementation Schedule

Phase	Activity	Duration
1. Site survey & design	Roof and LV switchboard survey, detailed design, approvals	Weeks 1–3
2. Procurement	Ordering, shipping (Mombasa), customs clearing, delivery to site	Weeks 2–10
3. Installation	Ballast structure, panels, inverters, AC combiner, cabling & earthing	Weeks 10–13
4. Commissioning	Testing, monitoring set-up, grid synchronisation	Weeks 13–14
5. Training & handover	Operator training, as-built drawings and O&M manual	Week 14

11. Warranty & After-Sales Support

Component	Warranty
Solar panels	15 yrs product / 30 yrs linear power
Inverters	5 years (extendable)
Mounting structure	10 years
Installation workmanship	2 years

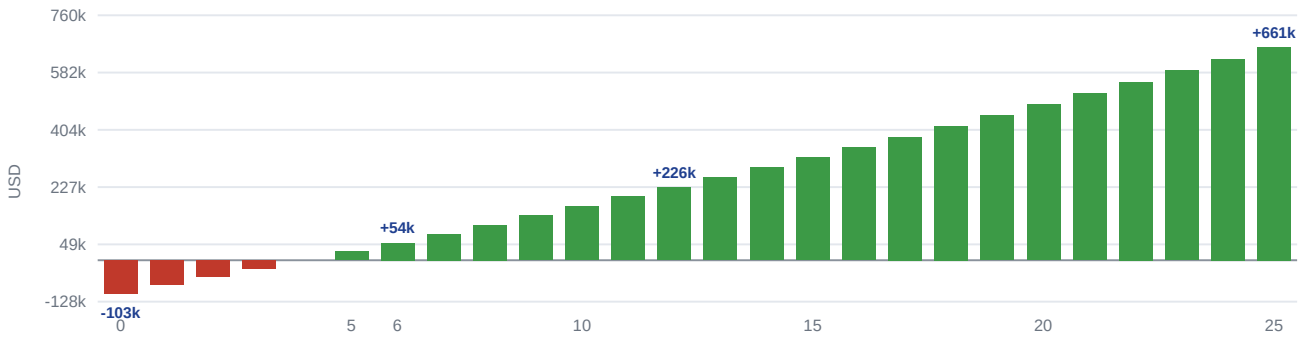
Installed to IEC solar installation standards and Uganda Electricity Regulatory Authority requirements. After handover we provide remote monitoring support, a free 6-month check-up visit and optional annual maintenance (panel cleaning, connection checks, inverter software updates, performance report).

12. Financial Analysis & Cash Flow

Assumption	Value
Turnkey investment (excl. VAT)	USD 102,789
Avoided tariff, year 1	USD 0.0871/kWh (shoulder)
Tariff escalation	2 % per year
Degradation	≈ 0.4 % per year
O&M cost / discount rate	none assumed / 10 %

Result	Value
Year-1 savings	USD 25,343
Simple payback	4.0 years
IRR (25 years)	25.9 %
NPV @ 10 %	USD 154,964
LCOE (solar energy cost)	USD 0.040/kWh
25-year savings	USD 763,309

Cumulative cash flow (USD) – the investment is recovered during year 4 (4.0 years).



Year	Generation (kWh)	Tariff (USD/kWh)	Savings (USD)	Net cash flow (USD)	Cumulative (USD)
0				(102,789)	(102,789)
1	291,051	0.087	25,343	25,343	(77,446)
3	285,840	0.091	25,895	25,895	(26,063)
5	283,558	0.094	26,726	26,726	26,970
6	282,424	0.096	27,151	27,151	54,121
7	281,294	0.098	27,583	27,583	81,704
10	277,932	0.104	28,922	28,922	167,116
12	275,713	0.108	29,850	29,850	226,349
15	272,418	0.115	31,298	31,298	318,780
20	267,013	0.127	33,870	33,870	482,906
25	261,715	0.140	36,654	36,654	660,520

No O&M or inverter-replacement cost is included. Values in brackets are negative (investment not yet recovered).

13. Quotation for 185.4 kWp Rooftop Solar PV – Turnkey EPC

Quote No.	NSP-261105-KMH	Date	05-Nov-2026	Valid Until	05-Dec-2026
Client	Kampala Marriott Hotel	Contact	Mahil Fernando	Currency	USD

#	Scope of supply & works	Qty	Unit	Price (USD)
1	PV modules – Astronergy ASTRO N7 620 Wp bifacial N-type TOPCon	299	nos	**USD 102,789**
2	String inverters – Sigenergy Sigen PV 80M1, 80 kW, 1100 V	2	nos	
3	Ballasted aluminium mounting structure, 10° tilt	1	lot	
4	DC system – PV cable, connectors, string isolators, cable tray	1	lot	
5	AC system – AC combiner panel, MCCBs, cabling to main LV switchboard	1	lot	
6	Smart data logger, energy meter & remote monitoring	1	set	
7	Earthing & lightning protection integration	1	lot	
8	Installation, testing & commissioning	1	lot	
9	Transport, logistics & customs clearing to site	1	lot	
	TOTAL TURNKEY EPC PRICE (excluding VAT and other applicable taxes)			**USD 102,789**

Equivalent to USD 0.55 per Wp. Amount in words: One hundred two thousand seven hundred eighty-nine US Dollars only.

Payment terms	75 % advance with PO · 25 % on commissioning
Delivery	Equipment 8–10 weeks from PO and advance; installation & commissioning ≈ 4 weeks after delivery
Taxes	VAT, withholding tax and statutory levies excluded – charged as applicable
Exclusions	Roof strengthening / waterproofing, battery storage, utility and permit fees, works not listed above
Client to provide	Roof access, connection to the existing NW10 H1 solar incomer on the main LV switchboard, power & water during installation

14. Financial Summary & Payback

USD 102,789 Total investment (excl. VAT)	**USD 25,343** Savings in year 1	**4.0 years** Payback period	**25.9 %** IRR (25 years)
USD 763,309 25-year savings	**USD 660,520** 25-year net gain after investment	**7.4 x** Savings / investment	**USD 154,964** NPV @ 10 %

15. Tariff Sensitivity

Avoided tariff (USD/kWh)	Year-1 savings (USD)	Payback (years)	IRR	NPV @ 10 % (USD)
0.0522	15,206	6.5	15.7 %	51,863
0.07	20,374	4.9	21.0 %	104,424
0.0871 (base)	**25,343**	**4.0**	**25.9 %**	**154,964**
0.10	29,105	3.5	29.6 %	193,230
0.1027	29,904	3.4	30.4 %	201,359
0.12	34,926	2.9	35.3 %	252,433

The project remains attractive over a wide tariff range. The break-even tariff (NPV = 0 at 10 %) is about **USD 0.0347/kWh** – roughly 60 % below today's shoulder tariff. Other rows: with the 40 % energy rebate = USD 0.0522; VAT not recoverable = USD 0.1027.

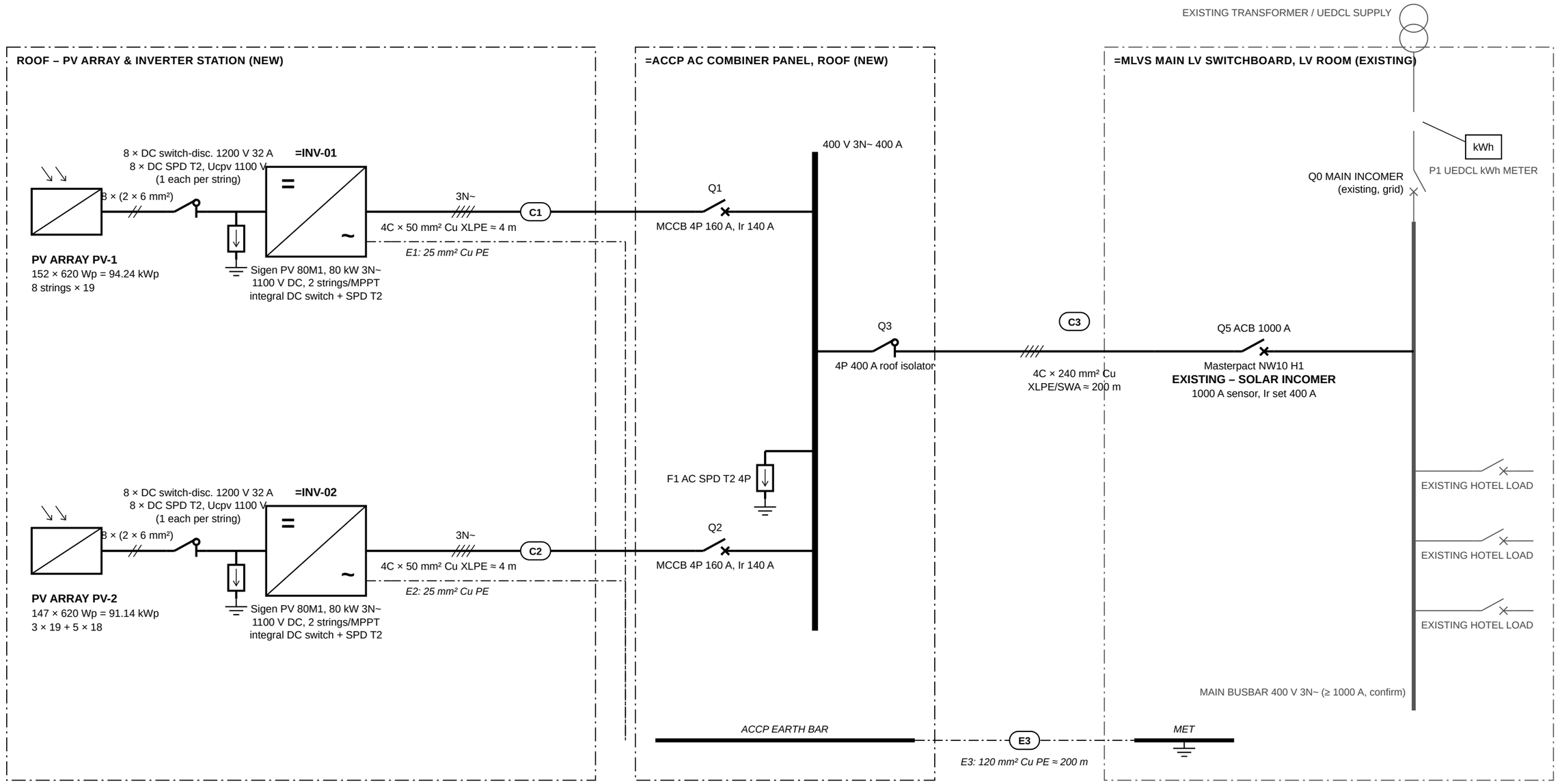
Annexes

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Annex 2	Solar PV layout – mezzanine roof	11
Annex 3	Data sheet – Sigenergy Sigen PV inverter (80 kW)	12–13
Annex 4	Data sheet – Astronergy ASTRO N7 620 W bifacial solar module	14–15
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Annex 6	Quotation – 185.4 kWp rooftop solar PV (one page)	18
Annex 7	PVsyst simulation report – P50 energy yield (V8.1.8, VC0)	19–25

Next Steps

1. Review of this proposal and clarification meeting with the hotel's engineering team.
2. Site survey: roof condition, LV switchboard, cable route and inverter location.
3. Confirmation of the tariff from recent utility bills and structural review of the roof.
4. Purchase order and advance payment – project start.

Thank you for considering NileSun Power Ltd. Contact: Mahil Fernando | +256 708669574 | sales@nilesunpower.com



SYMBOLS (IEC 60617)

- PV array / PV generator
- Inverter DC/AC
- Circuit breaker (MCCB / ACB)
- Switch-disconnector (isolator)
- Surge protective device (SPD)
- Earth / main earthing terminal
- Heavy = new, light = existing; strokes = conductors
- DC conductor · PE conductor

CABLE SCHEDULE

TAG	FROM → TO	CABLE	LENGTH	PROTECTION
S1–S16	PV strings → inverters	2 × 6 mm² Cu H1Z2Z2-K	≈ 40 m	DC isolator 32 A + DC SPD T2
C1, C2	INV-01/02 → ACCP Q1/Q2	4C × 50 mm² Cu XLPE	≈ 4 m	MCCB 160 A (Ir 140 A)
C3	ACCP Q3 → MLVS Q5	4C × 240 mm² Cu XLPE/SWA	≈ 200 m	ACB NW10 H1, Ir 400 A
E1, E2	INV-01/02 → ACCP earth bar	25 mm² Cu (G/Y)	≈ 4 m	–
E3	ACCP earth bar → MET	120 mm² Cu (G/Y)	≈ 200 m	–
–	Array bonding / spine	16 mm² / 35 mm² Cu (G/Y)	≈ 280 m	–

Lengths indicative – confirm on site. Sizes per IEC 60364-5-52 / 5-54 (see design report).

NOTES

1. 299 × 620 Wp = 185.4 kWp DC; 2 × 80 kW = 160 kW AC; DC/AC 1.16.
2. Grid-tied, no battery. DC SPD T2 + DC isolator on every string.
3. C3 200 m: VD ≈ 2.0 % at 232 A. Q5 = existing NW10 H1, Ir 400 A.
4. Final ratings to equipment datasheets and detailed design.

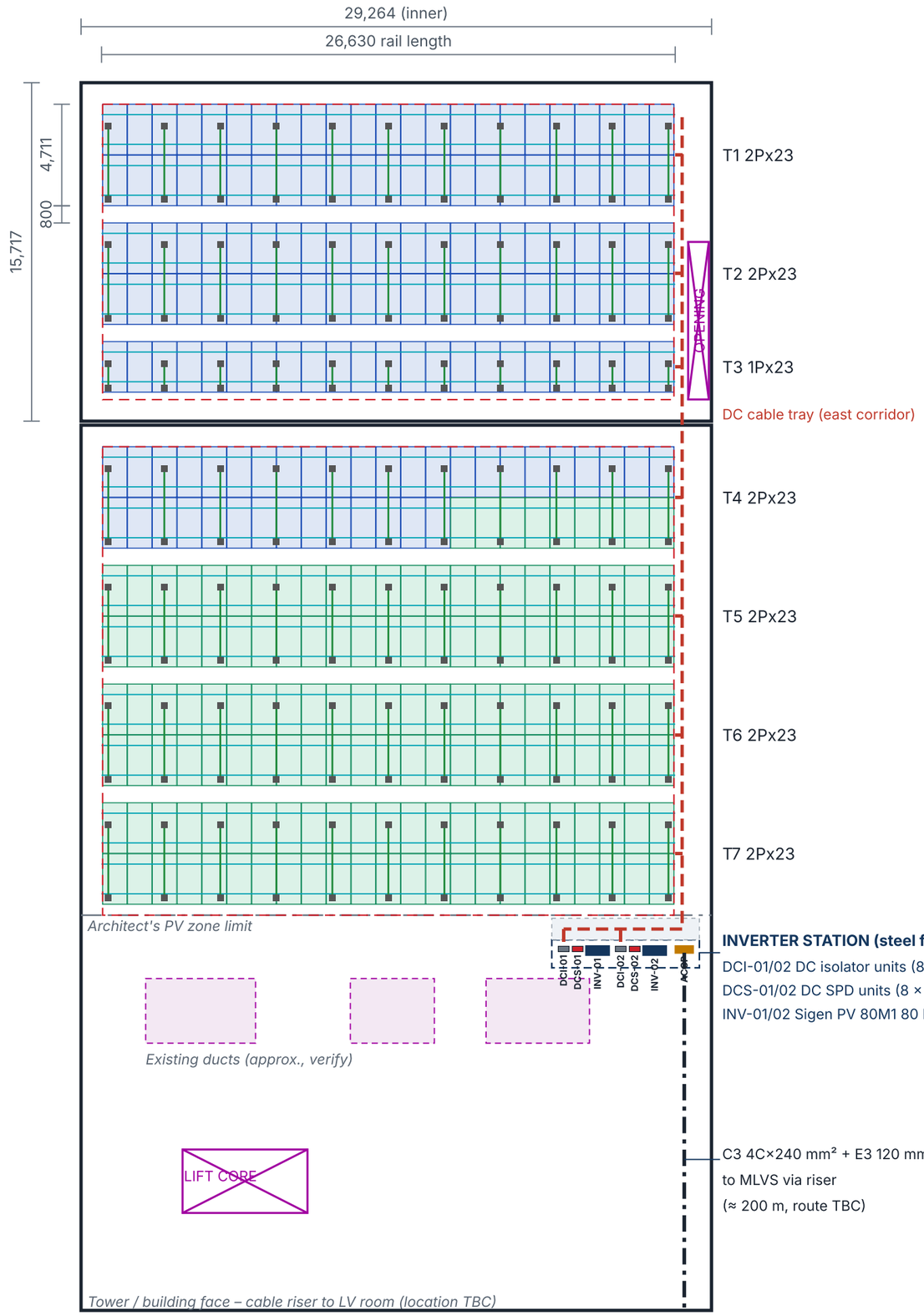
CLIENT: **Kampala Marriott Hotel**	PROJECT: **Kampala Marriott – Rooftop PV 185.4 kWp**

TITLE: **SINGLE-LINE DIAGRAM – PV SYSTEM & GRID CONNECTION**

DRAWING No.	REV	SCALE / SIZE	DATE
PV-SLD-D18102-004	**B**	**NTS / A3**	**09.11.2026**

DRAWN	CHECKED	APPROVED

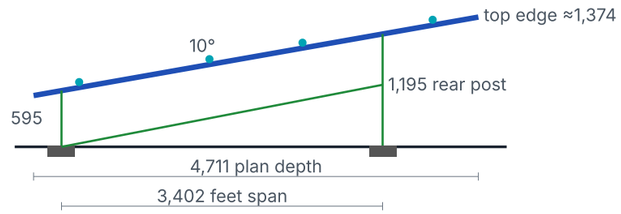
PLAN – PV LAYOUT, INVERTERS & ACCP ON MEZZANINE ROOF (ENERACK BALLAST, 10° TILT)



INVERTER STATION (steel frame + canopy)
DCI-01/02 DC isolator units (8 × 2P 1200 V 32 A each)
DCS-01/02 DC SPD units (8 × Type 2, 1100 V each)
INV-01/02 Sigen PV 80M1 80 kW · ACCP 400 A

C3 4C×240 mm² + E3 120 mm²
to MLVS via riser
(≈ 200 m, route TBC)

SECTION A – 2P TABLE (2 × 23 modules)



SECTION B – 1P TABLE (1 × 23 modules)

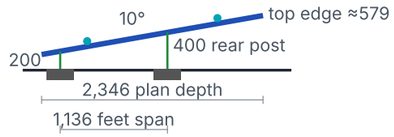


TABLE SCHEDULE

Table	Type	Modules	Inverter
T1, T2	2P × 23	92	INV-01
T3	1P × 23	23	INV-01
T4	2P × 23	46	INV-01: 37 · INV-02: 9
T5–T7	2P × 23	138	INV-02
TOTAL (77 frames, 26 rail lines)		**299**	**INV-01 152 · INV-02 147**

DESIGN DATA

Module	620 Wp bifacial TOPCon, 2382 × 1134 × 30 mm
Structure	Enerack ballast, dwg EPJ-26110401 V1.0
Tilt / wind / snow	10° / 35 m/s / 0 kN/m²
Rails	R80, 8 × 3000 + 2630 = 26,630 mm per line
Frames	11 per table: first at 315, then 10 × 2,600
Spacing	20 mm module gap · 800 mm aisle · ≥ 950 mm rail to wall

LEGEND

INV-01 modules · INV-02 modules · Inverter · ACCP · DC isolator unit · DC SPD unit
DC cable tray · C3 + E3 to LV room · Existing ducts
Rail · Frame / post · Foot block (300 × 300 indicative)
Setback line · Obstruction

NOTES: 1. Dimensions in mm. 2. Roof outline traced from D18102-DD-A-104; verify on site. 3. Foot block size and ballast weight per Enerack calculation; roof load to be approved by the building structural engineer. 4. All tables lie within the architect's solar zone. 5. Inverter station and duct positions indicative – confirm on site with the hotel; C3 riser route and LV room location TBC.

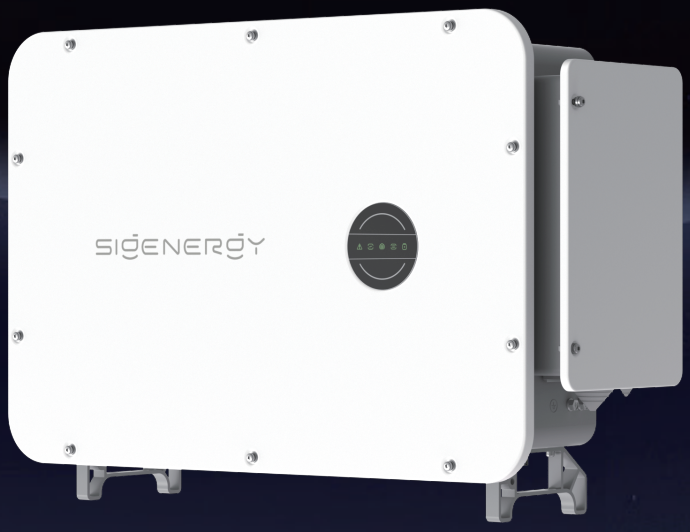
ROOFTOP SOLAR PV LAYOUT

Project	Marriott Hotel – Mezzanine Roof
Client	Kampala Marriott Hotel
Drawing No.	PV-LAY-D18102-002 Rev C
Date / Scale	9 November 2026 · NTS (A3)

SIGENERGY

Sigen PV Inverter

50.0 / 60.0 / 80.0 / 100.0 / 110.0 / 125.0 kW



- Smaller and lighter, easier installation and transportation
- Built-in EMS, supports 100 units in parallel without data logger
- Industry-leading 500m AFCI, top-tier safety across applications
- On-site self-power supply, removes the need for temporary power
- IP66 protection rating, ensuring worry-free outdoor deployment

Enjoy Green Energy

Sigen PV Inverter 50.0 / 60.0 / 80.0 / 100.0 / 110.0 / 125.0 kW

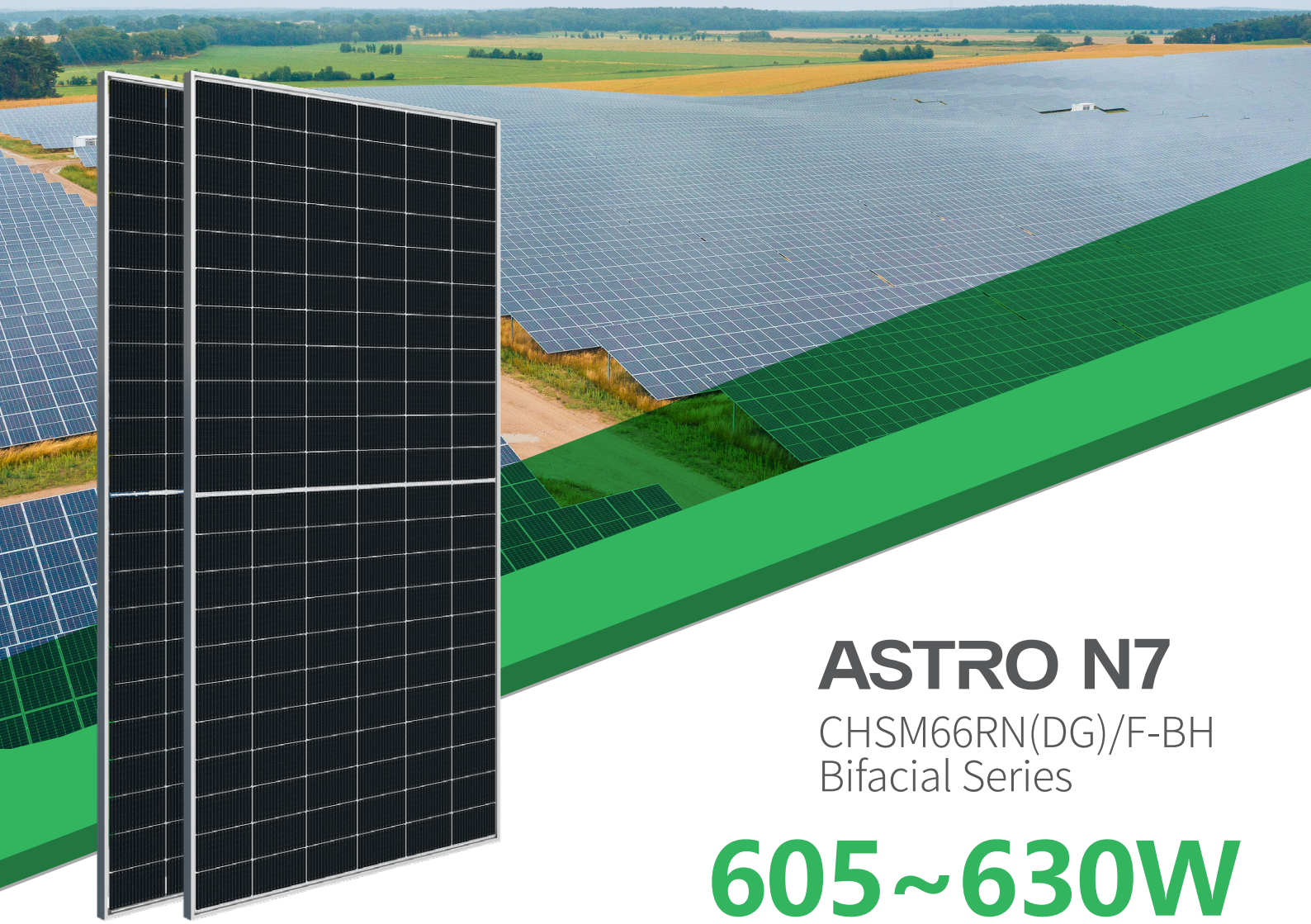
Sigen PV	50M1	60M1	80M1	100M1	110M1	125M1	Units
DC Input							
Max. PV input power	100,000	120,000	160,000	200,000	220,000	220,000	Wp
Max. DC input voltage	1,100						V
Nominal DC input voltage	600						V
Start-up voltage	180						V
MPPT voltage range	160 ~ 1,000						V
Number of MPP. trackers	4	5	6	8	8	8	
Number of PV strings per MPPT	2						
Max. input current per MPPT	40						A
Max. short-circuit current per MPPT	60						A
AC Output							
Nominal output active power	50,000	60,000	80,000	100,000	110,000	125,000	W
Max. output apparent power	55,000	66,000	88,000	110,000	121,000	137,500	VA
Max. output active power (cosΦ=1)	55,000	66,000	88,000	110,000	121,000	137,500	W
Nominal output current @380Vac	76.0	91.2	121.5	151.9	167.1	189.9	A
Nominal output current @400Vac	72.5	87.0	115.9	144.9	159.4	181.2	A
Max. output current @380 / 400Vac	83.6	100.3	133.7	167.1	183.8	208.9	A
Nominal output voltage	380 / 400, 3W+(N)+PE						Vac
Nominal grid frequency	50 / 60						Hz
Power factor	0.8 leading ~ 0.8 lagging						
Total current harmonic distortion	THDi < 3%	THDi < 3%	THDi < 2%	THDi < 2%	THDi < 2%	THDi < 2%	
Efficiency							
Max. efficiency	98.6%						
European efficiency	98.3%	98.3%	98.3%	98.4%	98.4%	98.3%	
Protection							
Safety protection feature	DC reverse polarity protection, Insulation monitoring, Residual current monitoring, Arc fault circuit interrupter, AC overcurrent/overvoltage/short-circuit protection. Type II DC/AC surge protection, Anti-islanding protection						
General Data							
Dimensions (W / H / D)	918 / 640 / 340						mm
Weight	75	75	75	75	75	82	kg
Nighttime power consumption	< 3.5						W
Storage temperature range	-40 ~ 70						°C
Operating temperature range	-30 ~ 60						°C
Relative humidity range	0% ~ 100%						
Max. operating altitude	5,000 (Derating at 4,000m)						m
PV connection type	MC4 (Max. 6 mm²)						
AC connection type	OT / DT terminal (Max. 240 mm²)						
Cooling	Smart air cooling						
Ingress protection rating	IP66						
Communication	WLAN / Fast Ethernet / RS485 / Sigen CommMod (4G/3G/2G)						
Standard Compliance							
Standard [1]	IEC / EN 62109-1, IEC / EN 62109-2, IEC / EN 61000-6-1, IEC / EN 61000-6-2						

1. For all standards refer to the certificates category on the Sigenergy website.
2. The information in this document reflects the current state of technology and is subject to change without notice. For the latest updates, please refer to the Sigenergy website.

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Doc. Version 20250418

ASTRONERGY



ASTRO N7

CHSM66RN(DG)/F-BH
Bifacial Series

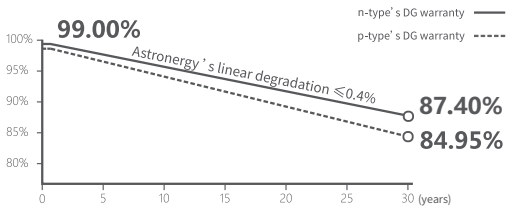
605~630W

Assist carbon neutrality

Warranty

15 15-year Product Warranty

30 30-year Linear Power Warranty



n-type TOPCon 4.0
Novel upgrade, enhancing module efficiency

SMBB Design
Enhancing current collection, minimizing power loss

Low Voc Design
Increasing modules per string, lower BOS cost

Bifacial Power Generation
Maximizing bifaciality, boosting backside power output

IEC 61215, IEC 61730
ISO 9001:2015:ISO Quality Management System
ISO 14001:2015:ISO Environment Management System
ISO 45001:Occupational Health and Safety
The first solar company which passed the Nord IEC/TS 62941 certification audit

TOP PERFORMER 2024 kiwa PVEL PV MODULE RELIABILITY SCORECARD

Tier 1 BloombergNEF

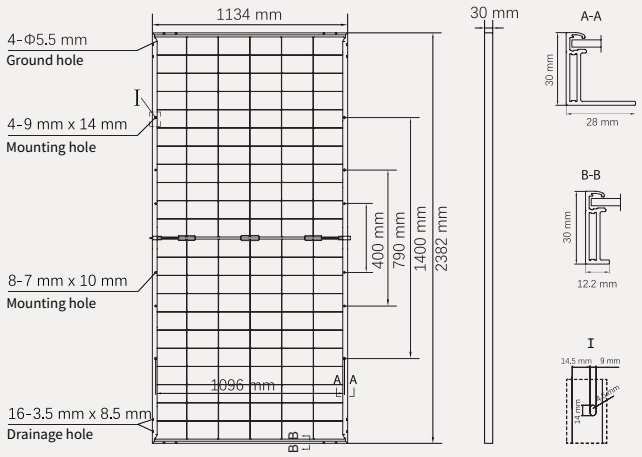
Underwritten by International Insurer

605~630W	0~+3%	23.3%	≤ 1.0%	≤ 0.4%
POWER RANGE	POWER SORTING	MAX MODULE EFFICIENCY	FIRST YEAR POWER DEGRADATION	YEAR 2-30 POWER DEGRADATION

Mechanical Specifications

Outer dimensions (L x W x H)	2382 x 1134 x 30 mm
Cell type	n-type mono-crystalline
No. of cells	132 (6*22)
Frame technology	Aluminum, silver anodized
Front / Back glass	2.0+2.0 mm
Cable length (Including connector)	Portrait: (+)350 mm,(-)250 mm; Customized length
Cable diameter (IEC/UL)	4 mm^2 / 12 AWG
① Maximum mechanical test load	5400 Pa (front) / 2400 Pa (back)
Connector type (IEC/UL)	HCB40 (Standard) / MC4-EVO2A (Optional)
Module weight	32.8 kg
Packing unit	36 pcs / box
Weight of packing unit (for 40'HQ container)	1231 kg
Modules per 40' HQ container	720 pcs (Subject to sales contract)

① Refer to Astronergy crystalline installation manual or contact technical department.
Maximum Mechanical Test Load=1.5×Maximum Mechanical Design Load.



Electrical Specifications

STC: Irradiance 1000W/m^2, Cell Temperature 25° C, AM=1.5

Rated output (Pmpp / Wp)	605	610	615	620	625	630
Rated voltage (Vmpp / V)	41.18	41.31	41.43	41.56	41.69	41.82
Rated current (Impp / A)	14.69	14.77	14.84	14.92	14.99	15.07
Open circuit voltage (Voc / V)	48.59	48.74	48.89	49.04	49.19	49.34
Short circuit current (Isc / A)	15.86	15.94	16.02	16.11	16.19	16.27
Module efficiency	22.4%	22.6%	22.8%	23.0%	23.1%	23.3%

NMOT: Irradiance 800W/m^2, Ambient Temperature 20° C, AM=1.5, Wind Speed 1m/s

Rated output (Pmpp / Wp)	455.0	458.7	462.5	466.2	470.0	473.8
Rated voltage (Vmpp / V)	38.76	38.88	39.00	39.12	39.24	39.36
Rated current (Impp / A)	11.74	11.80	11.86	11.92	11.98	12.04
Open circuit voltage (Voc / V)	46.15	46.30	46.44	46.58	46.72	46.86
Short circuit current (Isc / A)	12.80	12.87	12.94	13.00	13.07	13.13

Electrical Specifications (Integrated power)

Pmpp gain	Pmpp / Wp	Vmpp / V	Impp / A	Voc / V	Isc / A
5%	646	41.43	15.59	48.89	16.83
10%	677	41.43	16.33	48.89	17.63
15%	707	41.43	17.07	48.89	18.43
20%	738	41.43	17.81	48.89	19.23
25%	769	41.43	18.55	48.89	20.03

Electrical characteristics with different rear power gain (reference to 615W)

Temperature Ratings (STC)

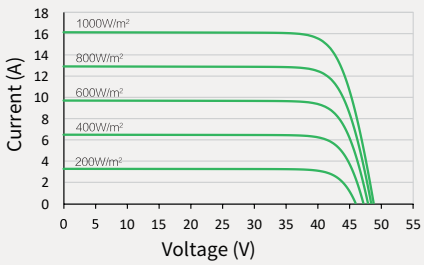
Temperature coefficient (Pmpp)	-0.29%/°C
Temperature coefficient (Isc)	+0.043%/°C
Temperature coefficient (Voc)	-0.25%/°C
Nominal module operating temperature (NMOT)	41±2°C

Operating Parameters

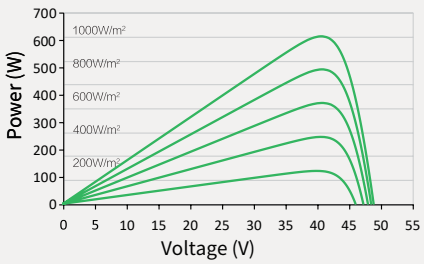
No. of diodes	3
Junction box IP rating	IP 68
Max. series fuse rating	35 A
Max. system voltage (IEC/UL)	1500V_{DC}

Curve

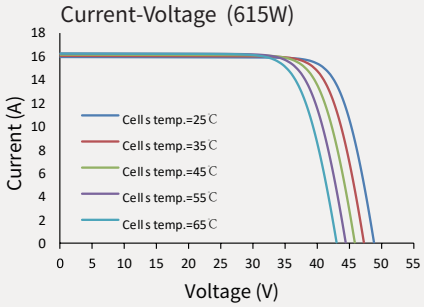
Current-Voltage (615W)



Power-Voltage (615W)

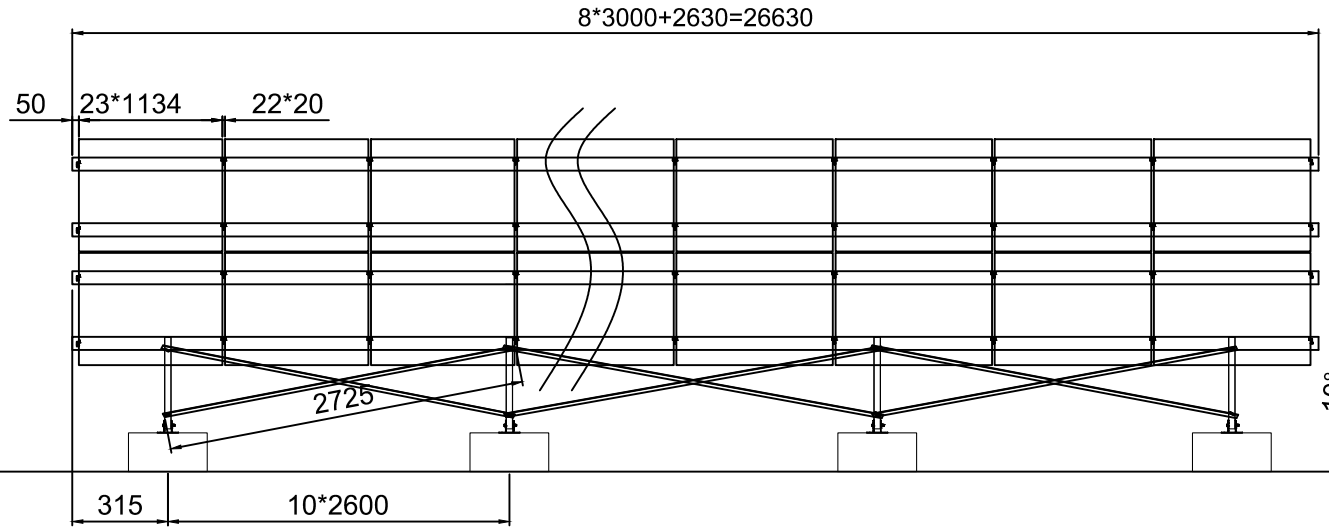


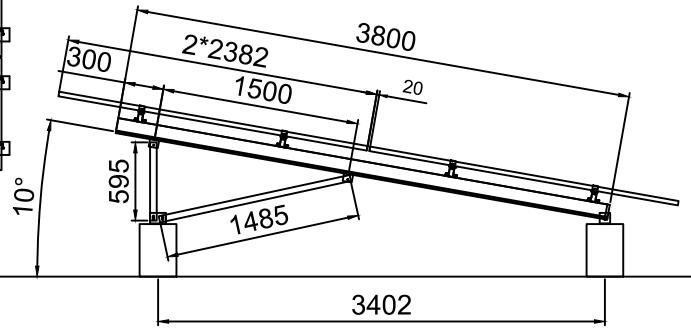
Current-Voltage (615W)

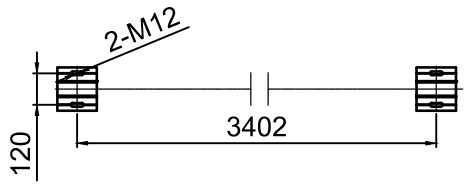


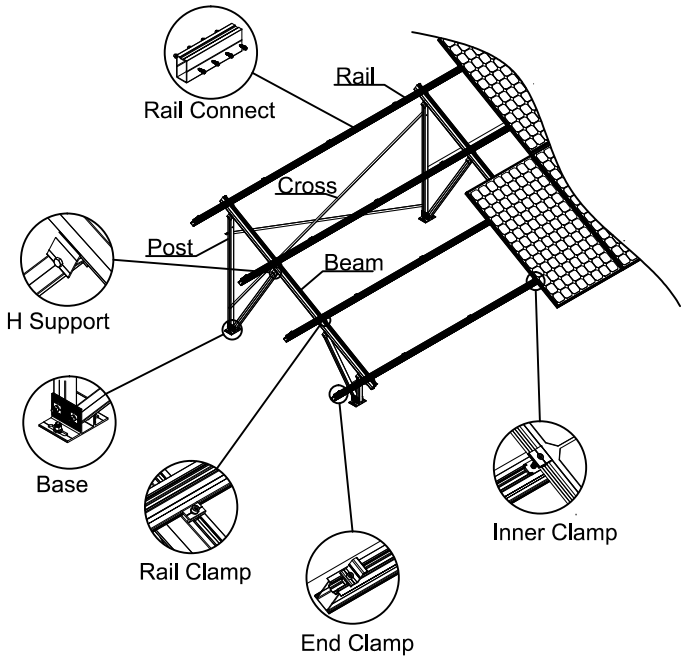
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www.astronergy.com

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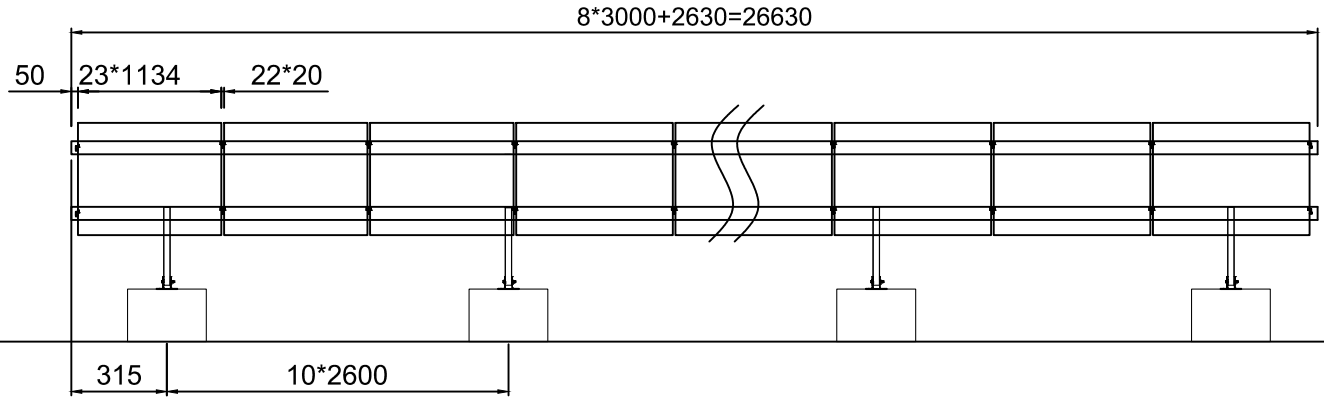


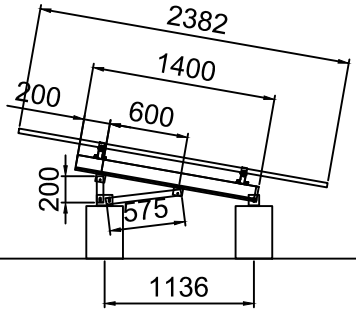
NOTES:
1.Panels Size: 2382 × 1134 × 30 mm
2.Wind Load: 35m/s
3.Snow Load: 0kN/m²
4.Install angle: 10°

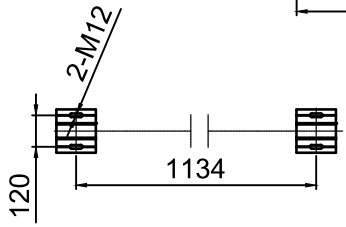
Dwg No.	EPJ-26110401	Array	2*24	Version	V1.0
Design		Chk'D		Appv'D	

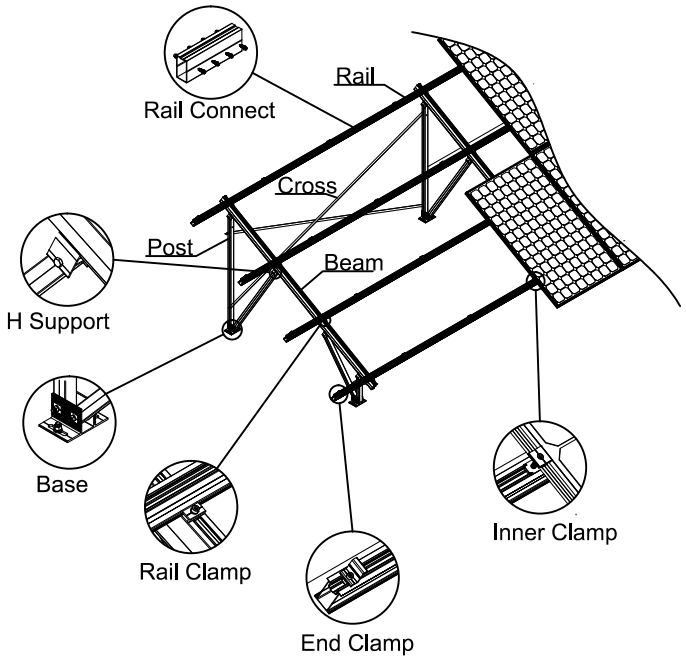
Xiamen Enerack Technology Co., Ltd.

View	
Scale	1:1
Unit	mm









NOTES:
1.Panels Size: 2382 × 1134 × 30 mm
2.Wind Load: 35m/s
3.Snow Load: 0kN/m²
4.Install angle: 10°

Dwg No.	EPJ-26110401	Array	2*23	Version	V1.0
Design		Chk'D		Appv'D	

Xiamen Enerack Technology Co., Ltd.

View	
Scale	1:1
Unit	mm

Nile Sun Power Limited
Kampala, Uganda

Commercial Proposal & Quotation – Rooftop Solar PV System

Turnkey EPC supply, installation and commissioning · 185.4 kWp grid-tied · no battery

Quotation No.:	NSP-261105-KMH	**Date:**	05-11-2026
Revision:	A	**Valid until:**	05-12-2026 (30 days)

1. Project Details

Project name:	Kampala Marriott Hotel – 185.4 kWp Rooftop Solar PV (mezzanine roof)
System:	Grid-tied solar PV, self-consumption, no battery storage
Location:	Kampala, Uganda · 0°18'11.34"N, 32°35'13.76"E · Ref. drawing D18102-DD-A-104

2. Client & Supplier Details

Client:	Kampala Marriott Hotel	**Supplier:**	Nile Sun Power Limited
Address:	Kampala, Uganda	**Address:**	Kampala, Uganda
Contact:	Mr. Jagath	**Contact:**	Mahil Fernando
Mobile / Email:	+256 791 811805	**Mobile / Email:**	+94 773 55 96 82 · mahil@windforce.lk

3. Quotation (Turnkey EPC)

No.	Scope of supply & works	Qty	Unit	Total Price (USD)
1	PV modules – Astronergy ASTRO N7 620 Wp bifacial N-type TOPCon	299	nos	**USD 102,789**
2	String inverters – Sigenergy Sigen PV 80M1, 80 kW, 1100 V	2	nos	
3	Ballasted aluminium mounting structure, 10° tilt	1	lot	
4	DC system – PV cable, connectors, string isolators, cable tray	1	lot	
5	AC system – AC combiner panel, MCCBs, cabling to main LV switchboard	1	lot	
6	Smart data logger, energy meter & remote monitoring	1	set	
7	Earthing & lightning protection integration	1	lot	
8	Installation, testing & commissioning	1	lot	
9	Transport, logistics & customs clearing to site	1	lot	
	TOTAL TURNKEY EPC PRICE (excluding VAT and other applicable taxes)			**USD 102,789**

Equivalent to USD 0.55 per Wp · Amount in words: One hundred two thousand seven hundred eighty-nine US Dollars only.

4. Payment & Commercial Terms

Price basis:	Turnkey EPC, delivered, installed and commissioned at site, Kampala. Currency: USD.
Payment terms:	75 % advance with purchase order · 25 % on commissioning
Taxes:	Prices exclude VAT, withholding tax and any other statutory levies, which shall be charged as applicable.

5. Delivery & Project Schedule

Programme:	Equipment delivery 8–10 weeks from PO and advance payment; installation and commissioning ≈ 4 weeks after delivery (≈ 12–14 weeks in total).

6. Warranties

PV modules:	15-yr product · 30-yr linear power	**Inverters:**	5 years (extendable)
Mounting structure:	10 years	**Workmanship:**	2 years from commissioning

7. Notes & Exclusions

1. Final quantities are subject to detailed engineering and site survey; price will be adjusted for any change in scope.
2. Excluded: roof structural strengthening or waterproofing works, battery storage, utility/permit fees, and works not listed above.
3. Client to provide roof access, connection to the existing Masterpact NW10 H1 solar incomer on the main LV switchboard, and power/water during installation.
4. Energy and savings estimates are given in the accompanying proposal and are indicative only.

PVsyst V8.1.8

PHOTOVOLTAIC SOFTWARE

PVsyst TRIAL

PVsyst - Simulation report

Grid-Connected System

Project: Kampala_Marriott_Hotel

Variant: Kampala_Marriott_Hotel

No 3D scene defined, no shadings

System power: 185 kWp

Nsambya - Uganda

Author

Project: Kampala_Marriott_Hotel

Variant: Kampala_Marriott_Hotel

PVsyst V8.1.8
VC0, Simulation date:
09/11/26 18:38
with V8.1.8

Project summary

Geographical Site
Nsambya
Uganda

Situation

Latitude	0.3042 °(N)
Longitude	32.5874 °(E)
Altitude	1178 m
Time zone	UTC+3

Project settings
Albedo 0.20

Weather data
Nsambya
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	299 units	Nb. of units	2 units
Pnom total	185 kWp	Total power	160 kWac
		Pnom ratio	1.16

Results summary

Simulation step	Hourly				
Produced Energy	291051 kWh/year	Specific production	1570 kWh/kWp/year	Perf. Ratio PR	85.44 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Main results 4
Loss diagram 5
Predef. graphs 6
Single-line diagram 7

PVsyst TRIAL

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VC0, Simulation date:
09/11/26 18:38
with V8.1.8

General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	CHSM66RN(DG)/F-BH-620 (Original PVsyst database)
Unit Nom. Power	620 Wp
Number of PV modules	299 units
Nominal (STC)	185 kWp
Modules	23 units x 13 In series
At operating cond. (50°C)	
Pmpp	172 kWp
U mpp	491 V
I mpp	351 A

Inverter

Manufacturer	Generic
Model	80M1-HYA (Original PVsyst database)
Unit Nom. Power	80.0 kWac
Number of inverters	2 units
Total power	160 kWac
Operating voltage	160-1000 V
Max. power (=>40°C)	88.0 kWac
Pnom ratio (DC:AC)	1.16
Power sharing within this inverter	

Total PV power

Nominal (STC)	185 kWp
Total	299 modules
Module area	808 m²
Cell area	754 m²

Total inverter power

Total power	160 kWac
Max. power	176 kWac
Number of inverters	2 units
Pnom ratio	1.16

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 23 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.05 %

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

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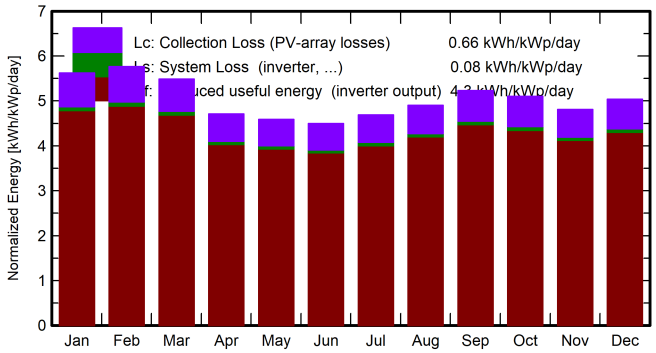
Main results

Simulation step
Hourly

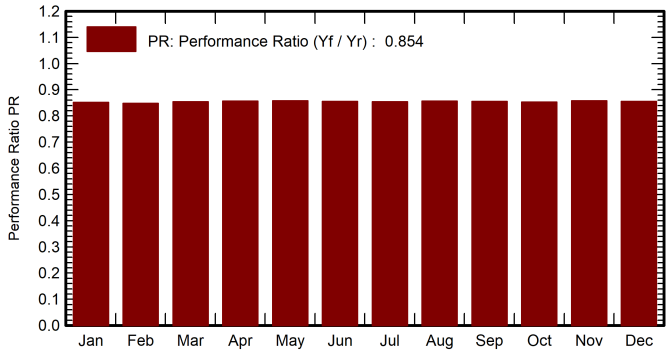
System Production

Produced Energy	291051 kWh/year	Specific production	1570 kWh/kWp/year
		Perf. Ratio PR	85.44 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

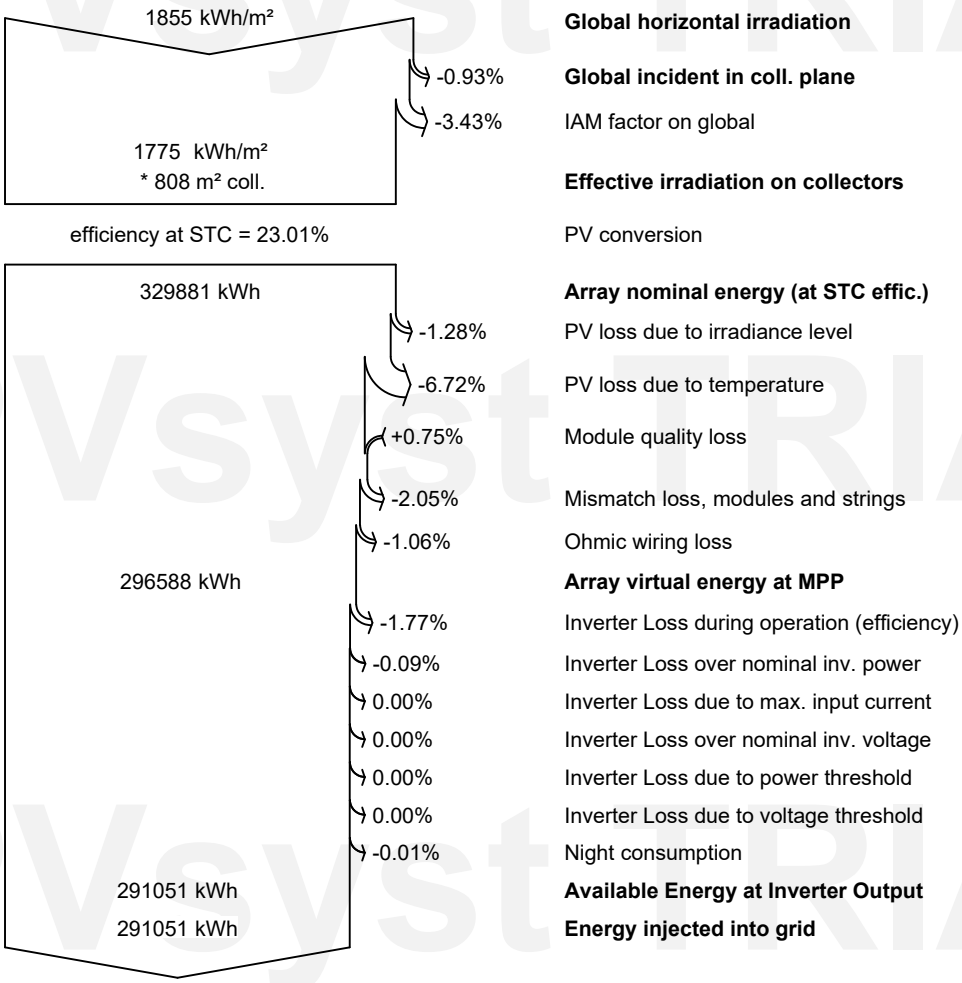
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	163.9	65.15	23.02	174.4	169.0	28018	27523	0.851
February	155.8	60.45	23.52	161.5	157.0	25848	25387	0.848
March	170.3	76.10	23.21	170.2	164.9	27424	26942	0.854
April	146.6	72.32	22.79	141.3	136.1	22846	22443	0.857
May	152.4	71.94	22.59	142.3	136.5	23031	22628	0.858
June	147.4	61.35	22.19	135.0	129.0	21771	21387	0.855
July	158.5	56.87	21.80	145.4	139.4	23454	23038	0.855
August	160.2	63.78	21.90	152.1	146.5	24591	24154	0.857
September	159.5	65.84	22.11	157.0	152.1	25340	24885	0.855
October	155.5	68.89	22.31	158.1	153.4	25458	24993	0.853
November	138.1	73.98	22.30	144.3	139.6	23357	22933	0.857
December	146.8	67.95	22.42	156.1	151.3	25187	24738	0.855
Year	1855.0	804.62	22.51	1837.7	1774.7	296323	291051	0.854

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram



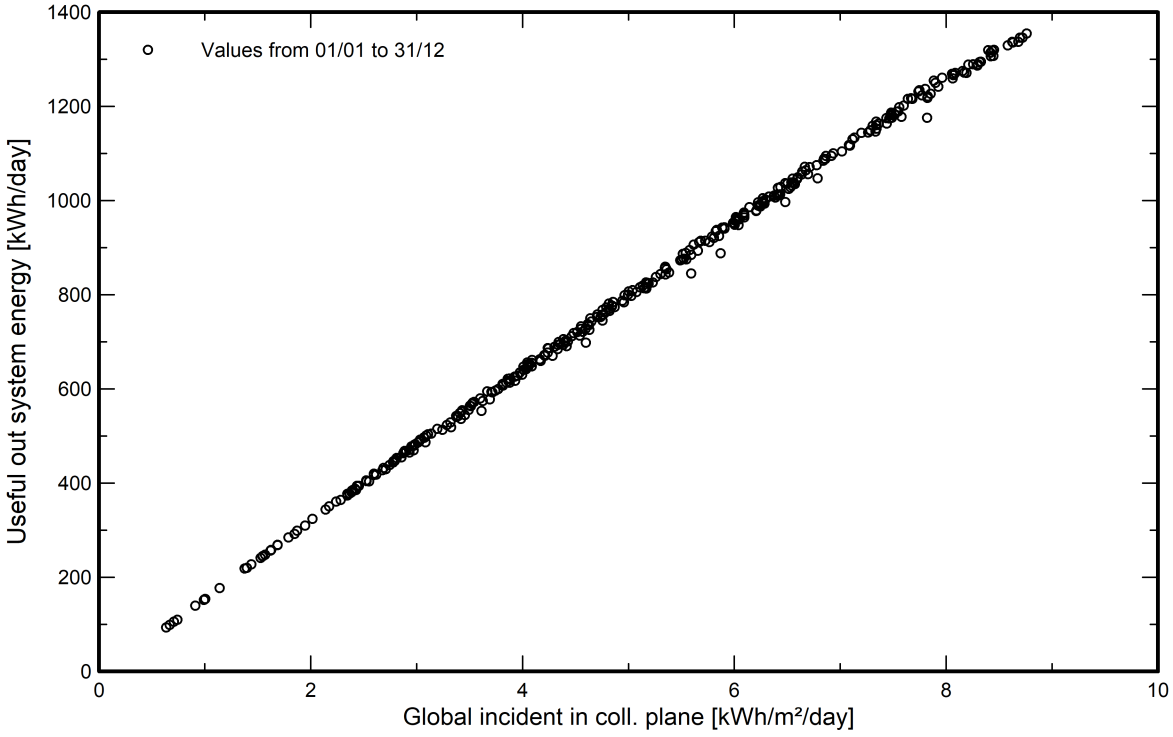
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with V8.1.8

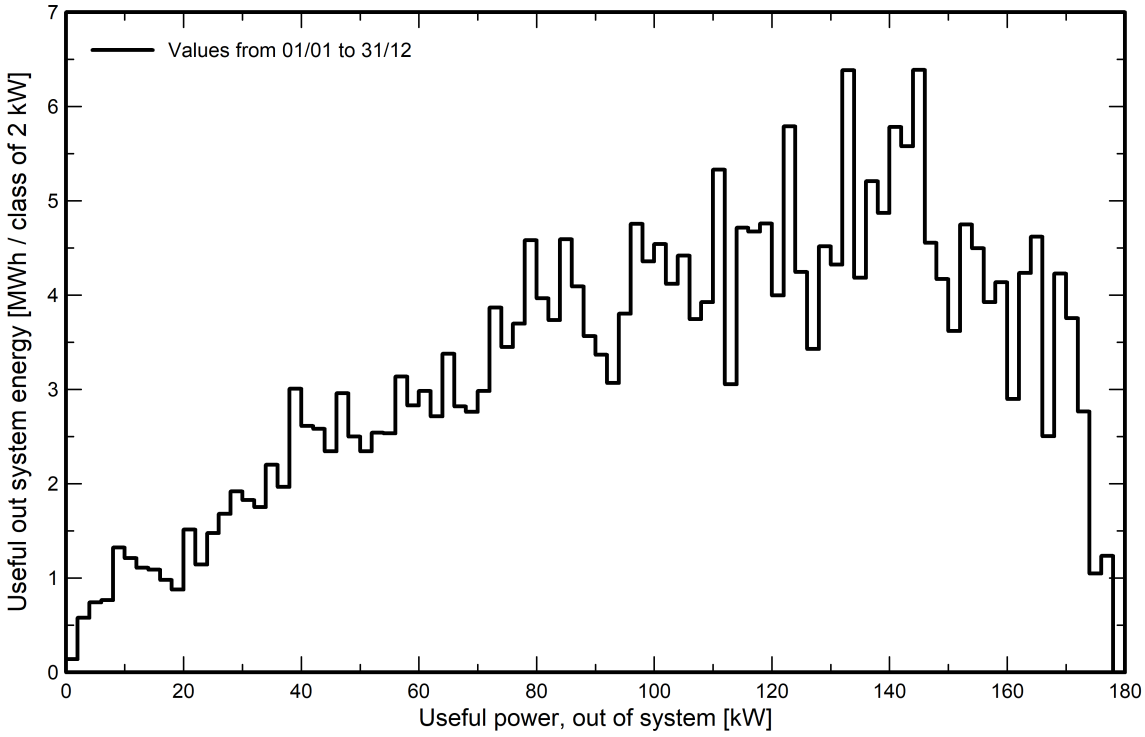
PVsyst TRIAL

Predef. graphs

Daily Input/Output diagram

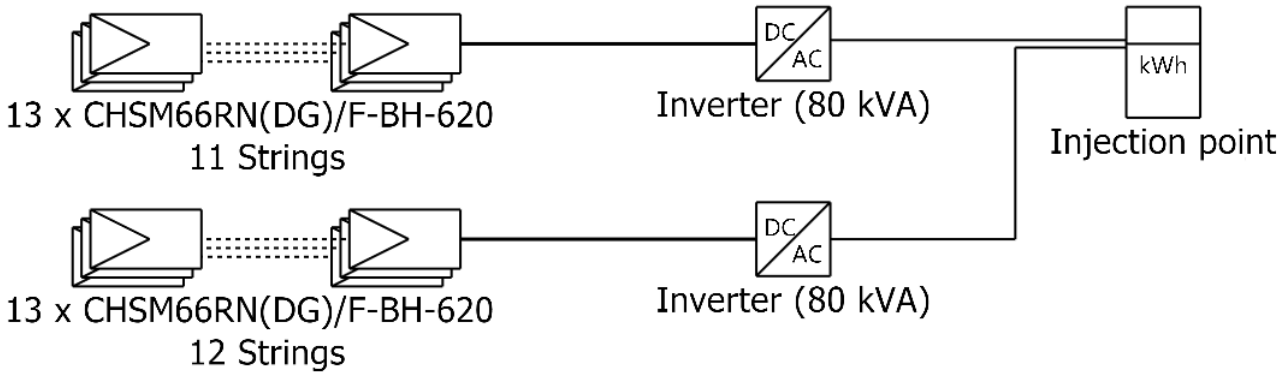


System Output Power Distribution



Single-line diagram

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09/11/26 18:38
with V8.1.8



PV module	CHSM66RN(DG)/F-BH-620
Inverter	80M1-HYA
String	13 x CHSM66RN(DG)/F-BH-620

	Nsambya_160_kW	
	VC0 : Nsambya_160_kW	09/11/26